Lecture 3 - Computer Hardware

Overview

This lecture introduces the basics of **computer components and architecture**, including the **CPU**, **buses, clocks, and registers**, **memory hierarchy**, and **I/O systems**. It also covers **primary and secondary storage**, **storage devices**, and the **role of the CPU** in executing instructions, providing a foundation for understanding computer operation.

Topics

1. **The Components of a Computer**
2. **CPU Basics**
3. **The Bus**
4. **Clocks**
5. **Registers**
6. **Main Memory (ROM vs RAM)**
7. **Primary and Secondary Storage**
8. **Storage Devices**

Objectives / Outcome

1. Identify and describe the main components of a **computer system**.
2. Understand the basics of the **CPU** and its role in computing.
3. Explain the function of the **bus**, **clocks**, and **registers** in a computer system.
4. Describe the different types of **computer memory** and the memory hierarchy.
5. Understand the **input/output subsystem** and the concepts of **memory-mapped** and **instruction-based I/O**.

The Components of a Computer

Overview

A computer system consists of several key components, each with a specific function.

Components

- **Central Processing Unit (CPU):** The brain of the computer that performs instructions.
- **Memory (RAM and ROM):** Stores data and instructions temporarily or permanently.
- **Storage Devices (HDD, SSD):** Used for long-term data storage.
- **Input Devices (Keyboard, Mouse):** Allow users to provide data and commands to the computer.
- **Output Devices (Monitor, Printer):** Present information from the computer to the user.
- **Motherboard:** Connects all components and allows communication between them.
- **Power Supply Unit (PSU):** Provides power to all computer components.
- **Graphics Processing Unit (GPU):** Handles rendering of images, video, and graphics.

CPU Basics

Central Processing Unit (CPU)

The brain of the computer that performs instructions defined by software. It consists of the **Arithmetic Logic Unit (ALU)**, **Control Unit (CU)**, and **registers**.

Components

- **ALU:** Performs arithmetic and logical operations.
- **CU:** Directs operations within the computer by fetching and decoding instructions.
- **Registers:** Small, fast storage locations used to hold data temporarily.

Example Question

Q: What are the main components of a CPU and their functions?
A: The main components of a CPU are the ALU (performs arithmetic and logical operations), CU (directs operations within the computer), and registers (hold data temporarily).

The Bus

Definition

A communication system that transfers data between components inside a computer or between computers.

Types

- **Data Bus:** Transfers actual data.
- **Address Bus:** Carries the address to which the data is being sent.
- **Control Bus:** Carries control signals from the CPU to other components.

Example Question

Q: What is the function of the address bus in a computer system?
A: The address bus carries the address to which the data is being sent.

Clocks

Clock

A device that generates a consistent signal used to synchronize operations in the computer.

Clock Speed

Measured in Hertz (Hz), it indicates the number of cycles per second. Higher clock speed means more instructions processed per second.

Example Question

Q: How does the clock speed of a CPU affect its performance?
A: Higher clock speed means more instructions can be processed per second, leading to better performance.

Registers

Definition

Small, fast storage locations within the CPU used to hold data temporarily.

Types

- **Accumulator:** Holds intermediate results of ALU operations.
- **Instruction Register (IR):** Holds the current instruction being executed.
- **Program Counter (PC):** Holds the address of the next instruction to be executed.

SUMMARY TABLE

FEATURE	Cache	Registers
DEFINITION	Small, high-speed memory for frequently accessed data	Small, fast storage within the CPU for immediate data processing
LOCATION	Close to the CPU, often on the CPU chip	Within the CPU itself
SPEED	Faster than main memory, slower than registers	Fastest type of memory
SIZE	Larger than registers, smaller than main memory	Very small, a few bytes
PURPOSE	Reduce access time to main memory	Provide immediate access to data and instructions
TYPES	L1, L2, L3 Cache	Accumulator, Instruction Register, Program Counter, General Purpose Registers
VOLATILITY	Volatile	Volatile
ACCESS METHOD	Managed by cache controller	Directly accessed by the CPU's control unit